Evaluate Yourself

7) 0.185 g of an organic compound when treated with Conc. HNO_3 and silver nitrate gave 0.320 g of silver bromide. Calculate the % of bromine in the compound. (Ag =108, Br = 80) Ans: 73.6

8) 0.40 g of an iodo-substituted organic compound gave 0.235 g of AgI by carius method. Calculate the percentage of iodine in the compound. (Ag = 108, I = 127) (Ans = 31.75)

Estimation of phosphorus:

Carius method: A known mass of the organic compound (w) containing phosphorous is heated with fuming HNO_3 in a sealed tube where C is converted into CO_2 and H to H_2O. phosphorous present in organic compound is oxidized to phosphoric acid which is precipitated, as ammonium phosphomolybdate by heating with Conc. HNO_3 and then adding ammonium molybdate.

$$H_3PO_4 + 12\,(NH_4)_2\,MoO_4 + 21\,HNO_3 \xrightarrow{\text{heat}} (NH_4)_3\,PO_4.12MoO_3 + 21NH_4\,NO_3 + 12\,H_2O$$

The precipitate of ammonium phosphomolybdate thus formed is filtered washed, dried and weighed.

In an alternative method, the phosphoric acid is precipitated as magnesium-ammonium phosphate by adding magnesia mixture (a mixture containing $MgCl_2$, NH_4Cl and ammonia) This ppt is washed, dried and ignited to get magnesium pyrophosphate which is washed, dried a weighed. The following are the reaction that takes place.

By knowing the mass of the organic compound and the mass of ammonium phosphomolybdate or magnesium pyrophosphate formed, the percentage of P is calculated.

Mass of organic compound is wg

Weight of ammonium

phosphomolybdate = x g

Weight of magnesium pyrophosphate = y g

Mole mass of $(NH_4)_3PO_4.12MoO_3$ is =1877g

[3 x(14 + 4) + 31 +4(16)] + 12 (96+3x16)

Molar mass of $Mg_2P_2O_7$ is 222 g

(2x24) + (31x2) + (7x16)

1877g of $(NH_4)_3PO_4.12MoO_3$ contains 31g of P

xg of $(NH_4)_3PO_4.12\ MoO_3$ in w g of organic compound contains $\frac{31}{1877}\times\frac{x}{w}$ of phosphorous

Percentage of Phosphorous $=\frac{31}{1877}\times\frac{x}{w}\times 100\%$

(or)

222 of $Mg_2P_2O_7$ contains 62 g of P

Y g of of $Mg_2P_2O_7$ in w g of Organic compound contains $\frac{62}{222}\times\frac{y}{w}$ of P.

Percentage Phosphorous $=\frac{62}{222}\times\frac{y}{w}\times 100\%$

Example 4: 0.24 g of organic compound containing phosphorous gave 0.66 g of $Mg_2P_2O_7$ by the usual analysis. Calculate the percentage of phosphorous in the compound

Weight of an organic compound = 0.24 g

Weight of $Mg_2P_2O_7$ = 0.66 g

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